CASE AND SCENARIO		STIMULUS FROM ./emu → REPLY (-r) · EXPECTED PINS P7–P4 (-o), IN ORDER
C01	Normal temperature and current	25.4 A · 26.5 °C `A5 · 00` → STATUS `007F3540`
C02	Warm boundaries: 27.0 and 32.0 °C	25.4 A · 27.0 °C `A5 · 10` → 25.4 A · 32.0 °C `A5 · 10` → STATUS `007F4041`
C03	Temperature Critical	25.4 A · 32.5 °C `A5 · A0` → STATUS `017F414B`
C04	Current boundaries: 25.6 and 32.0 A	25.4 A · 25.0 °C `A5 · 00` → 25.6 A · 25.0 °C `A5 · 40` → 32.0 A · 25.0 °C `A5 · 40`
C05	Current Critical (trip)	32.2 A · 25.0 °C `A5 · C0` → STATUS `01A1324C`
C06	Either fault is enough	32.2 A · 32.5 °C `A5 · E0` → 32.2 A · 25.0 °C `A5 · C0` → 25.4 A · 32.5 °C `A5 · A0`
C07	Both must become Normal	32.2 A · 32.5 °C `A5 · E0` → 25.6 A · 27.0 °C `A5 · D0` → 25.4 A · 26.5 °C `A5 · 80` → P0 pressed `00 cleared`
C08	Warm temperature blocks recovery	25.4 A · 32.5 °C `A5 · A0` → 25.4 A · 32.0 °C `A5 · 90` → 25.4 A · 26.5 °C `A5 · 80` → P0 pressed `00 cleared`
C09	Warning current blocks recovery	32.2 A · 25.0 °C `A5 · C0` → 25.6 A · 25.0 °C `A5 · C0` → P0 pressed `C0 refused` → 25.4 A · 25.0 °C `A5 · 80` → P0 pressed `00 cleared`
C10	Invalid packets change nothing	25.4 A · 32.5 °C `A5 · A0` → temperature raw 0 `3C · A0` → corrupted CRC `5A · A0`
C11	P0 refused until temperature and current are safe	25.4 A · 32.5 °C `A5 · A0` → P0 pressed `A0 refused` → 25.4 A · 27.0 °C `A5 · 90` → P0 pressed `90 refused` → 25.4 A · 26.5 °C `A5 · 80` → P0 pressed `00 cleared`
C12	A held P0 never clears the EPO	P0 held down `—` → 32.2 A · 25.0 °C `A5 · C0` → 25.4 A · 25.0 °C `A5 · 80` → release, press again `00 cleared`

`00` all off / EPO cleared · `10` P4 fan boost, `40` P6 current warning · `80` P7 EPO (+ `20` fans max, `10`, `40`) · reply bytes (`-r`)

Samples at 230 V unless stated. A5 accepted, 3C invalid, 5A bad CRC; each P0 press is followed by a release (`-i 0x00`).